PRANEEL SHAH

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EDUCATION

University of California, Berkeley | B.S. Chemical Biology & Bioengineering *Aug 2025 - May 2029*

Monta Vista High School *Aug 2021 - Jun 2025*

Relevant Coursework: Chem 4A, Math 52, Bioengineering 10, AP Chemistry, AP Biology, AP Calculus AB, AP Physics 1, AP Statistics
Awards & Honors: Regeneron Science Talent Search Top 300, Broadcom MASTERS 2nd in Technology Category, California Science & Engineering Fair Honorable Mention, PricewaterhouseCoopers Case Competition Top 12 Team in North America, Two Provisional Patents

PROFESSIONAL EXPERIENCE

University of California at Berkeley: Landry Lab, Research Assistant *Sep 2025 - Present*

- Collaborated with graduate mentors to study the mechanisms of how cell penetrating peptides (CPPs) enter the cell in cell-walled algal strains.
- Maintained sterile algal cultures and prepared specialized growth media to ensure long-term growth of cultures.
- Performed molecular biology workflows such as transformations and agarose electrophoresis to verify protein expression.

Nanoleaf Agriculture: Sustainable Farming Using Nanotechnology, Founder *May 2023 - Present*

- Founded a for-profit venture focused on commercializing innovative biocompatible nano-scaled fertilization and water filtration technologies.
- Developed and engineered high-performance prototypes, resulting in two patent-pending technologies.
- Beta-tested technology with 20+ households and a community garden, and currently partnered with NGOs to initiate large-scale pilot testing.

Green2Gold: Nonprofit Business Incubator, Pro-Bono Consultant *Jan 2023 - Present*

- Led outreach initiatives that expanded Green2Gold's incubator network to 400+ nonprofits and 1,000+ stakeholders.
- Advised NGOs within the incubator on scaling strategies and community engagement, supporting Green2Gold's mission of sustainability.
- Volunteered 500+ hours resulting in a Gold Presidential Service Award.

Massachusetts Institute of Technology (MIT), Research Assistant *Jun 2024 - Aug 2024*

- Designed and built an interactive research tool with Streamlit to analyze how 500+ gas molecule's spectra are distinguishable from each other.
- Implemented advanced data processing techniques, including spectral normalization, binning, and peak detection, to enable functional group labeling and suppress strong Q-branch features for improved spectral feature comparison.
- The application was featured in Professor Sara Seager's Fall 2024 Astrobiology class at MIT.

Aspiring Scholars Directed Research Program, Biochemistry Researcher *Aug 2022 - Feb 2024*

- Led the synthesis of iron nanoparticles to treat Parkinson's disease, and conducted extensive research using fourier-transform infrared spectroscopy, ultraviolet–visible spectroscopy, zeta potential, and scanning electron microscope characterizations.
- Completed 200+ lab hours, with manuscripts submitted to the National High School Journal of Science.

LEADERSHIP EXPERIENCE

Varsity & Club Volleyball (Mountain View Volleyball Club), Floor Captain *Jan 2024 - Jul 2025*

- Appointed Floor Captain junior and senior year, motivating teammates, mentoring underclassmen, and managing in-game situations.
- Guided team to two Central Coast Section Championships (2022 & 2024) and a Northern California State Championship Appearance (2024).
- Competed at AAU Boys' Junior Nationals (2021), finishing in the Top 5 nationally.

NASA Space Settlement Design Competition, Director of Operations *Jul 2023 - Jul 2024*

- Directed a 60+ person team of engineers and business strategists, combining design and business goals into a 50-page lunar settlement proposal.
- Orchestrated critical-path scheduling and dispute resolution, ensuring proposal delivery under a 24-hour deadline.
- Led final pitch to NASA engineers and aerospace executives, earning 1st place (2023) and Runner-Up (2024).

SKILLS & INTERESTS

Languages: Fluent in English, Hindi, Gujurati, and learning French
Software: Python, R-Programming, Java, Javascript, Procreate, Excel, Adobe Illustrator, Adobe Photoshop
Activities: Research, Volleyball, Green2Gold, Space Settlement Competitions, Data Analysis, Product Design
Interests: Chemistry, Sustainability, Science Fiction, Traveling, Music, Sports